Fund Transfer Requirements

Business Requirements Document (BRD)

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Data Notice	All values, accounts, names, and incidents in this document are fabricated for training. No production banking da

Revision History

Version	Date	Author	Change Summary
1.0	15-Mar-2024	R. Al Suwaidi	Initial baseline: BR-001 to BR-010.
2.0	08-Nov-2024	R. Al Suwaidi	Added OTP flow detail; introduced error code catalogue.
2.5	20-Apr-2025	M. Fernandes	Added idempotency (BR-011) after duplicate-posting incident INC-20250412-07.
3.0	10-Jan-2026	M. Fernandes	Added beneficiary cooling period (BR-012), cut-off rule (BR-013), sanction screening (BR-014),
3.1	03-Mar-2026	R. Al Suwaidi	Clarified that BR-005 OTP threshold is evaluated on the transfer amount only, excluding fees.
3.2	12-Jun-2026	R. Al Suwaidi	Daily limit now resets at 00:00 Gulf Standard Time, not UTC (see DEF-1007).

1. Purpose and Scope

This document defines the business rules that govern outbound fund transfers initiated by retail customers through digital channels (mobile app and internet banking). It is the authoritative source for validation logic implemented in the Fund Transfer Service and for the acceptance criteria used by the QA regression suite.

In scope: own-account transfers, within-bank transfers to a registered beneficiary, domestic transfers over local clearing, and international transfers over SWIFT.

Out of scope: cheque clearing, corporate bulk file uploads, standing instructions, credit card payments, and cash withdrawals. These are covered by separate BRDs.

2. Definitions

Term	Definition
Single Transfer	One transfer instruction submitted in one request. Fees are excluded when evaluating limits.
Daily Limit	Cumulative value of all successful outbound transfers by one customer within a calendar day, measured from 00:00:00
Available Balance	Ledger balance minus holds, minus unposted authorisations, plus approved overdraft limit where an overdraft facility is
Beneficiary	A payee registered against the customer profile. Holds a status of ACTIVE, PENDING, SUSPENDED, or DELETED.
OTP	One-Time Password delivered by SMS to the customer's registered mobile number. Valid for 180 seconds.
Frozen Account	An account with a debit block applied by Compliance, Legal, or Collections. Status code FRZ.
Transaction ID	A globally unique reference issued by the Fund Transfer Service in the format TRX-YYYYMMDD-XXXXXXXX.
Value Date	The business date on which the transfer is settled with the beneficiary bank.

3. Business Rules

3.1 Summary Matrix

ID	Rule	Priority	Error Code
BR-001	Maximum single transfer is AED 50,000.	Must	TRF-4001
BR-002	Maximum daily transfer amount is AED 100,000.	Must	TRF-4002
BR-003	Customers cannot transfer more than available balance.	Must	TRF-4003
BR-004	Beneficiary status must be ACTIVE.	Must	TRF-4004
BR-005	Transfers above AED 25,000 require OTP authentication.	Must	TRF-4005 / 4006 / 4007
BR-006	Amount must be greater than zero.	Must	TRF-4008
BR-007	Source and destination accounts cannot be identical.	Must	TRF-4009
BR-008	Currency must be supported by the account.	Must	TRF-4010
BR-009	Frozen accounts cannot initiate transfers.	Must	TRF-4011
BR-010	Successful transfer must generate a unique transaction ID.	Must	n/a
BR-011	Every transfer request must be idempotent for 24 hours.	Must	TRF-4012
BR-012	Newly added beneficiaries are subject to a 30-minute cooling period.	Must	TRF-4013
BR-013	Same-day domestic transfers must be submitted before the 21:00 GST cut-off.	Should	TRF-4014
BR-014	Transfers of AED 20,000 or above are screened against sanctions lists.	Must	TRF-4015
BR-015	Every accepted or rejected transfer must produce an audit record.	Must	n/a

3.2 Detailed Rules

BR-001 - Maximum Single Transfer Amount

A single transfer instruction must not exceed AED 50,000.00. The check is applied to the transfer amount only; correspondent fees and charges are excluded. For non-AED transfers the amount is converted to AED at the prevailing card rate before the check is applied.

#	Acceptance Criteria
AC1	Amount = 50,000.00 AED is ACCEPTED (boundary is inclusive).
AC2	Amount = 50,000.01 AED is REJECTED with TRF-4001.
AC3	A USD 20,000 transfer converting to AED 73,460 is REJECTED with TRF-4001.
AC4	The rejection message must state the limit and the customer's remaining daily allowance.

Linked test cases: TC-FT-001, TC-FT-002, TC-FT-003

BR-002 - Maximum Daily Transfer Amount

The cumulative value of all successful outbound transfers for one customer must not exceed AED 100,000.00 within a calendar day. The counter resets at 00:00:00 GST (UTC+4). Reversed and failed transfers must not consume the daily allowance. Pending transfers awaiting sanction clearance DO consume the allowance until they are rejected.

#	Acceptance Criteria
AC1	Customer has transferred AED 95,000 today; a new AED 5,000 transfer is ACCEPTED.
AC2	Customer has transferred AED 95,000 today; a new AED 5,001 transfer is REJECTED with TRF-4002.

#	Acceptance Criteria
AC3	A transfer that failed with TRF-5001 must be excluded from the daily total.
AC4	At 00:00:01 GST the counter must read 0.00.

Linked test cases: TC-FT-004, TC-FT-005, TC-FT-006, TC-FT-031

BR-003 - Available Balance Check

The debit amount plus all applicable fees must be less than or equal to the available balance of the source account at the moment of posting. The balance must be re-verified inside the posting transaction, not only at validation time, to prevent race conditions between concurrent requests.

#	Acceptance Criteria
AC1	Balance AED 10,000, transfer AED 10,000, fee AED 0 -> ACCEPTED.
AC2	Balance AED 10,000, transfer AED 10,000, fee AED 26.25 -> REJECTED with TRF-4003.
AC3	Two concurrent transfers of AED 6,000 against a balance of AED 10,000: exactly one succeeds.
AC4	An overdraft account with a limit of AED 5,000 and balance AED 1,000 may transfer up to AED 6,000.

Linked test cases: TC-FT-007, TC-FT-008, TC-FT-009, TC-FT-032

BR-004 - Beneficiary Status

The destination beneficiary must exist against the customer profile and hold status ACTIVE. Beneficiaries in PENDING, SUSPENDED, or DELETED status must be rejected. The response must not disclose which of the three non-active states applies.

#	Acceptance Criteria
AC1	Beneficiary status ACTIVE -> ACCEPTED.
AC2	Beneficiary status PENDING -> REJECTED with TRF-4004.
AC3	Beneficiary status SUSPENDED -> REJECTED with TRF-4004.
AC4	Beneficiary belonging to a different customer -> REJECTED with TRF-4004, not 403.

Linked test cases: TC-FT-010, TC-FT-011, TC-FT-012

BR-005 - OTP Authentication Threshold

Any transfer with an amount strictly greater than AED 25,000.00 requires successful OTP verification before posting. The OTP is valid for 180 seconds and allows a maximum of three verification attempts, after which the transfer is abandoned and a new instruction must be created. The threshold is evaluated on the transfer amount excluding fees (clarified in v3.1).

#	Acceptance Criteria
AC1	Amount = 25,000.00 -> NO OTP required.
AC2	Amount = 25,000.01 -> OTP required; posting without a verified OTP returns TRF-4005.
AC3	OTP submitted after 181 seconds -> TRF-4007.
AC4	Third consecutive wrong OTP -> transfer moves to ABANDONED and returns TRF-4006 with attemptsRemaining = 0.
AC5	An OTP issued for transfer A must not validate transfer B.

Linked test cases: TC-FT-013, TC-FT-014, TC-FT-015, TC-FT-016, TC-FT-033

BR-006 - Positive Amount

The amount must be a positive decimal value greater than zero, with at most two decimal places. Zero, negative, non-numeric, null, and values with more than two decimals must be rejected before any downstream call.

#	Acceptance Criteria
AC1	Amount = 0 -> REJECTED with TRF-4008.
AC2	Amount = -100 -> REJECTED with TRF-4008.
AC3	Amount = 100.123 -> REJECTED with TRF-4008.
AC4	Amount = "abc" -> REJECTED with HTTP 400 and TRF-4008.

Linked test cases: TC-FT-017, TC-FT-018, TC-FT-019

BR-007 - Distinct Source and Destination

The source account number and the destination account number must not be identical. The comparison is made on the normalised IBAN, ignoring case and whitespace.

#	Acceptance Criteria
AC1	Source AE070331234567890123456, destination identical -> REJECTED with TRF-4009.
AC2	Same IBAN written with spaces -> still REJECTED with TRF-4009.
AC3	Two different accounts held by the same customer -> ACCEPTED as an own-account transfer.

Linked test cases: TC-FT-020, TC-FT-021

BR-008 - Supported Currency

The transfer currency must be one of the currencies enabled on the source account. Supported currencies for retail accounts are AED, USD, EUR, GBP, and SAR. A currency not enabled on the account must be rejected even if the bank supports it globally.

#	Acceptance Criteria
AC1	AED transfer from an AED account -> ACCEPTED.
AC2	JPY transfer from an AED account -> REJECTED with TRF-4010.
AC3	USD transfer from an AED-only account -> REJECTED with TRF-4010.
AC4	Currency code in lower case ("aed") must be normalised and ACCEPTED.

Linked test cases: TC-FT-022, TC-FT-023, TC-FT-034

BR-009 - Frozen Account Restriction

An account carrying a debit block (status FRZ) must not initiate transfers. The check applies to the source account. A frozen destination account within the same bank must also be rejected. The customer-facing message must be generic and must not reveal the reason for the block.

#	Acceptance Criteria
AC1	Source account status FRZ -> REJECTED with TRF-4011.
AC2	Destination internal account status FRZ -> REJECTED with TRF-4011.
AC3	Account with a credit-only block may still receive but not send.
AC4	A transfer already OTP-verified must still be rejected if the account is frozen before posting.

Linked test cases: TC-FT-024, TC-FT-025, TC-FT-035

BR-010 - Unique Transaction Identifier

Every successful transfer must be assigned a globally unique transaction identifier in the format TRX-YYYYMMDD-XXXXXXXX, where XXXXXXXX is an eight-character uppercase alphanumeric sequence. The identifier must be returned in the API response, printed on the receipt, and used as the reconciliation key against the core banking ledger.

#	Acceptance Criteria
AC1	Two transfers created in the same millisecond receive different identifiers.
AC2	The identifier is returned in the 201 response body and in the Location header.
AC3	The identifier is immutable for the life of the transaction, including after reversal.
AC4	A reversal creates a NEW identifier that references the original in originalTransactionId.

Linked test cases: TC-FT-026, TC-FT-027, TC-FT-036

BR-011 - Idempotency

Every create-transfer request must carry an X-Idempotency-Key header. If the same key is replayed within 24 hours, the service must return the original result rather than creating a second transfer. A replay with the same key but a different request body must be rejected with TRF-4012.

#	Acceptance Criteria
AC1	Same key, same body, within 24h -> returns the original transaction with HTTP 200.
AC2	Same key, different amount -> REJECTED with TRF-4012.
AC3	Same key after 25 hours -> treated as a new request.
AC4	Missing header -> HTTP 400.

Linked test cases: TC-FT-028, TC-FT-037, TC-FT-038

BR-012 - New Beneficiary Cooling Period

A beneficiary added within the last 30 minutes may receive a maximum of AED 5,000 in aggregate. The cooling period starts from the beneficiary activation timestamp, not the creation timestamp.

#	Acceptance Criteria
AC1	Beneficiary added 10 minutes ago, transfer AED 5,000 -> ACCEPTED.
AC2	Beneficiary added 10 minutes ago, transfer AED 5,001 -> REJECTED with TRF-4013.
AC3	Beneficiary added 31 minutes ago, transfer AED 40,000 -> ACCEPTED.
AC4	Two transfers of AED 3,000 each inside the window: the second is REJECTED with TRF-4013.

Linked test cases: TC-FT-029, TC-FT-039

BR-013 - Same-Day Cut-off

Domestic transfers submitted after 21:00:00 GST, or on a UAE public holiday or a Sunday, are accepted but assigned the next business day as the value date. The customer must be shown the applicable value date before confirming.

#	Acceptance Criteria
AC1	Submitted 20:59 GST on a business day -> value date is today.
AC2	Submitted 21:01 GST -> value date is the next business day.
AC3	Submitted on Eid holiday -> value date is the next business day.
AC4	International SWIFT transfers use a separate 15:00 GST cut-off.

Linked test cases: TC-FT-030, TC-FT-040

BR-014 - Sanction Screening

Transfers with an amount of AED 20,000 or above, and all international transfers regardless of amount, must be submitted to the sanction screening service before posting. A screening hit places the transfer in status PENDING_COMPLIANCE for manual review and returns TRF-4015. Screening must complete within 4 seconds; a timeout must fail closed, not open.

#	Acceptance Criteria
AC1	Amount AED 19,999.99 domestic -> no screening call.
AC2	Amount AED 20,000.00 domestic -> screening call is made.
AC3	Any international transfer, even AED 100 -> screening call is made.
AC4	Screening service timeout -> transfer is held, never auto-approved.

Linked test cases: TC-FT-041, TC-FT-042

BR-015 - Audit Record

Every transfer attempt, whether accepted or rejected, must write an immutable audit record containing customer ID, timestamp in UTC, channel, device fingerprint, decision, and the rule ID that caused any rejection. Audit records are retained for 7 years. Audit writes must not block the customer response but must never be silently dropped.

#	Acceptance Criteria
AC1	A rejection under BR-004 writes an audit record naming BR-004.
AC2	The audit record contains no full PAN, no OTP value, and no password.
AC3	An audit write failure raises an operational alert within 60 seconds.
AC4	Audit records cannot be updated or deleted through any API.

Linked test cases: TC-FT-043, TC-FT-044

4. Transfer Type and Limit Matrix

Transfer Type	Channel	Single Limit (AED)	Daily Limit (AED)	OTP Threshold (AED)	Fee (AED)
Own Account	Mobile / Web	No limit	No limit	Not required	0.00
Within Bank	Mobile / Web	50,000	100,000	25,000	0.00
Domestic (local clearing)	Mobile / Web	50,000	100,000	25,000	1.05
Domestic (instant payment)	Mobile	10,000	20,000	5,000	0.50
International (SWIFT)	Web only	50,000	100,000	Always required	26.25

Note: Own-account transfers are exempt from BR-001, BR-002, and BR-005 but remain subject to BR-003, BR-006, BR-007, BR-008, BR-009, BR-010, and BR-015.

5. Non-Functional Requirements

ID	Requirement	Target
NFR-001	Transfer validation response time (p95)	under 800 ms
NFR-002	End-to-end posting response time (p95)	under 2.5 s
NFR-003	Service availability during business hours	99.95% monthly
NFR-004	Peak throughput (salary week)	450 transfers per second
NFR-005	Core banking call timeout and retry policy	6 s timeout, 2 retries, exponential backoff
NFR-006	OTP SMS delivery time (p90)	under 8 s
NFR-007	Audit record durability	no data loss, 7-year retention

6. Error Code Catalogue

Code	HTTP	Constant	Customer Message	Rule
TRF-4001	422	AMOUNT_EXCEEDS_SINGLE_LIMIT	The amount exceeds the single transfer limit of AED 50,000.	BR-001
TRF-4002	422	DAILY_LIMIT_EXCEEDED	You have reached your daily transfer limit.	BR-002
TRF-4003	422	INSUFFICIENT_FUNDS	Your available balance is not sufficient.	BR-003
TRF-4004	422	BENEFICIARY_NOT_ACTIVE	This beneficiary is not available for transfers.	BR-004
TRF-4005	401	OTP_REQUIRED	Please confirm this transfer with the code sent to your mobile.	BR-005
TRF-4006	401	OTP_INVALID	The code entered is incorrect.	BR-005
TRF-4007	401	OTP_EXPIRED	The code has expired. Please request a new one.	BR-005
TRF-4008	400	INVALID_AMOUNT	Please enter a valid amount.	BR-006
TRF-4009	422	SAME_ACCOUNT_TRANSFER	Source and destination accounts must be different.	BR-007
TRF-4010	422	CURRENCY_NOT_SUPPORTED	This currency is not available on the selected account.	BR-008
TRF-4011	403	ACCOUNT_RESTRICTED	This account cannot be used for transfers. Please contact the bank.	BR-009
TRF-4012	409	IDEMPOTENCY_CONFLICT	This request has already been processed.	BR-011
TRF-4013	422	BENEFICIARY_COOLING_PERIOD	New beneficiaries are limited to AED 5,000 for the first 30 minutes.	BR-012
TRF-4014	200	NEXT_DAY_VALUE_DATE	This transfer will be processed on the next business day.	BR-013

Code	HTTP	Constant	Customer Message	Rule
TRF-4015	202	COMPLIANCE_REVIEW	Your transfer is under review and will be processed shortly.	BR-014
TRF-5001	504	CORE_BANKING_TIMEOUT	We could not complete your request. Please try again.	NFR-005
TRF-5002	500	LEDGER_POSTING_FAILED	We could not complete your request. Please try again.	NFR-005

7. Traceability

Each business rule maps to at least one automated regression test in RegressionTestCases.xlsx and to the API contract published in BankingAPI-Swagger.json. Open defects against these rules are tracked in KnownDefects.csv. Field incidents are logged in ProductionIssues.txt.

Rule	API Operation	Test Cases	Open Defects
BR-001	POST /v1/transfers	TC-FT-001 to 003	-
BR-002	POST /v1/transfers	TC-FT-004 to 006, 031	DEF-1007
BR-003	POST /v1/transfers	TC-FT-007 to 009, 032	DEF-1002
BR-004	GET /v1/beneficiaries	TC-FT-010 to 012	DEF-1011
BR-005	POST /v1/transfers/{id}/otp/verify	TC-FT-013 to 016, 033	DEF-1001, DEF-1005
BR-006	POST /v1/transfers	TC-FT-017 to 019	-
BR-007	POST /v1/transfers	TC-FT-020, 021	DEF-1013
BR-008	POST /v1/transfers	TC-FT-022, 023, 034	DEF-1009
BR-009	POST /v1/transfers	TC-FT-024, 025, 035	DEF-1004
BR-010	POST /v1/transfers	TC-FT-026, 027, 036	DEF-1003
BR-011	POST /v1/transfers	TC-FT-028, 037, 038	DEF-1006
BR-012	POST /v1/transfers	TC-FT-029, 039	DEF-1015
BR-013	POST /v1/transfers	TC-FT-030, 040	DEF-1012
BR-014	POST /v1/transfers	TC-FT-041, 042	DEF-1008, DEF-1017
BR-015	internal audit sink	TC-FT-043, 044	DEF-1014